COMP4442 Service and Cloud Computing
Lab 2

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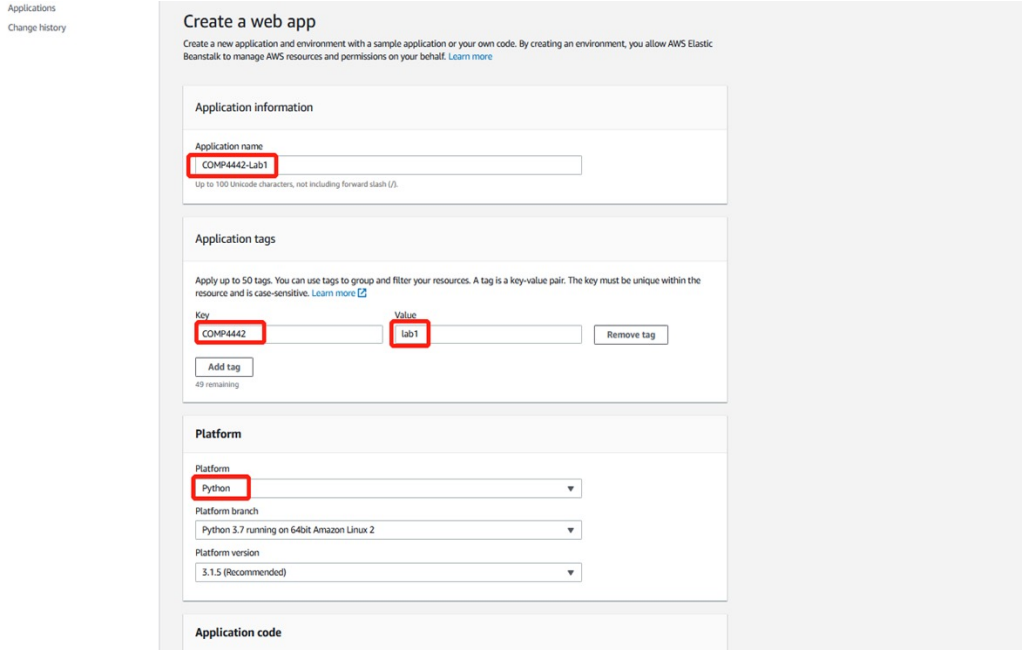
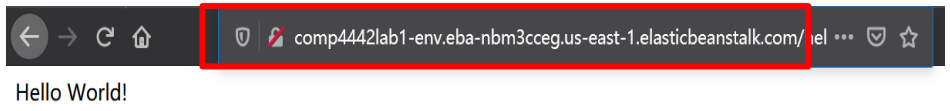
Outline

- Review of Lab-01
- Overview of AWS Simple Storage Service (S3)
- How to use AWS S3
  - Create bucket
  - Upload objects
  - Make objects available as URL
  - Move objects
  - Delete bucket and objects

Review of Lab-01

- We created AWS Education Account
- We learned how to use AWS services to develop a simple web application
  - Use Elastic Beanstalk with simple configuration

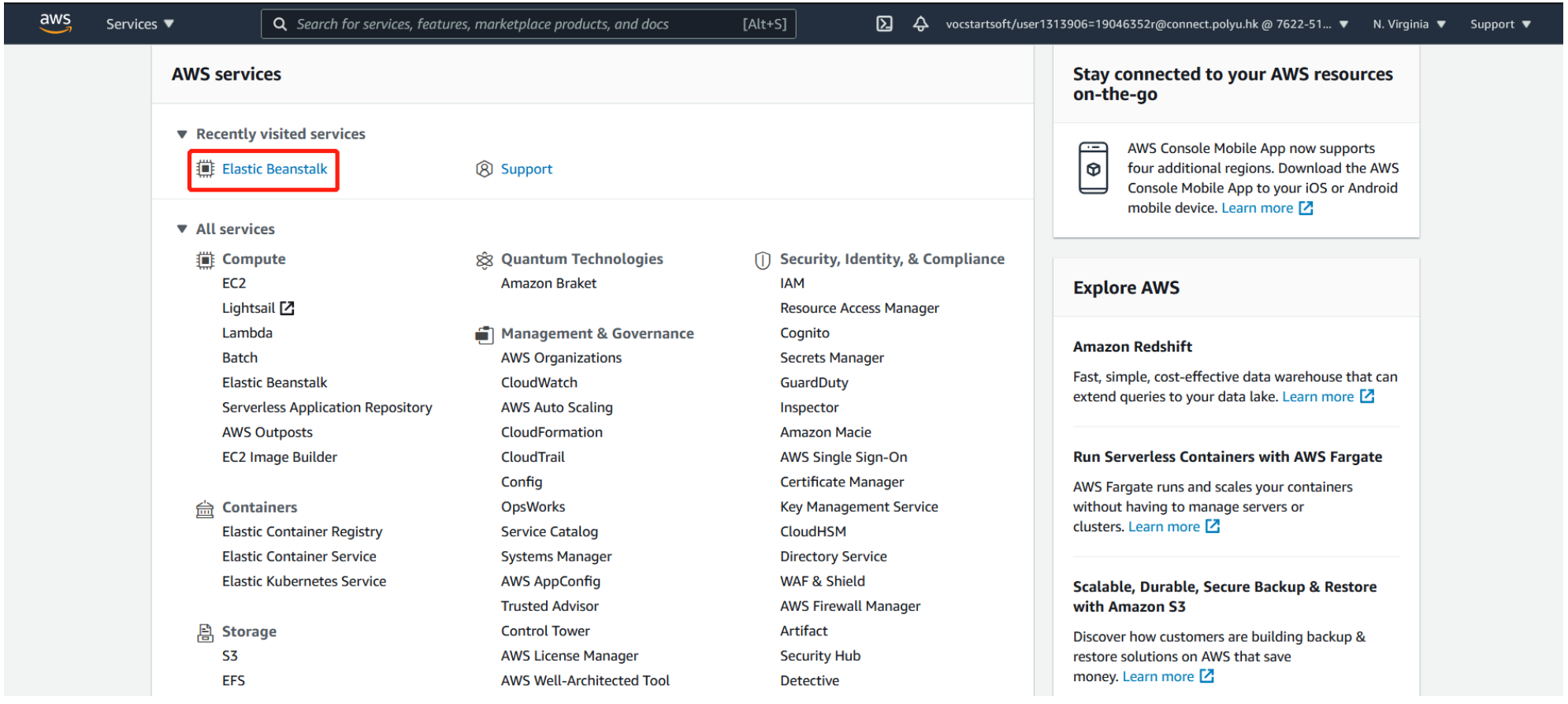




Hello World!

Stop Elastic Beanstalk

- Access the beanstalk



Stop Elastic Beanstalk

- Terminate the environment

aws Services Search for services, features, marketplace products, and docs [Alt+S] vocstartsoft/user1313906=19046352r@connect.polyu.hk @ 7622-51... N. Virginia Support

Elastic Beanstalk

Environments
Applications
Change history

Recent environments
Comp4442lab1-env

Elastic Beanstalk > Environments

All environments

Filter results matching the display values

2 Actions Create a new environment

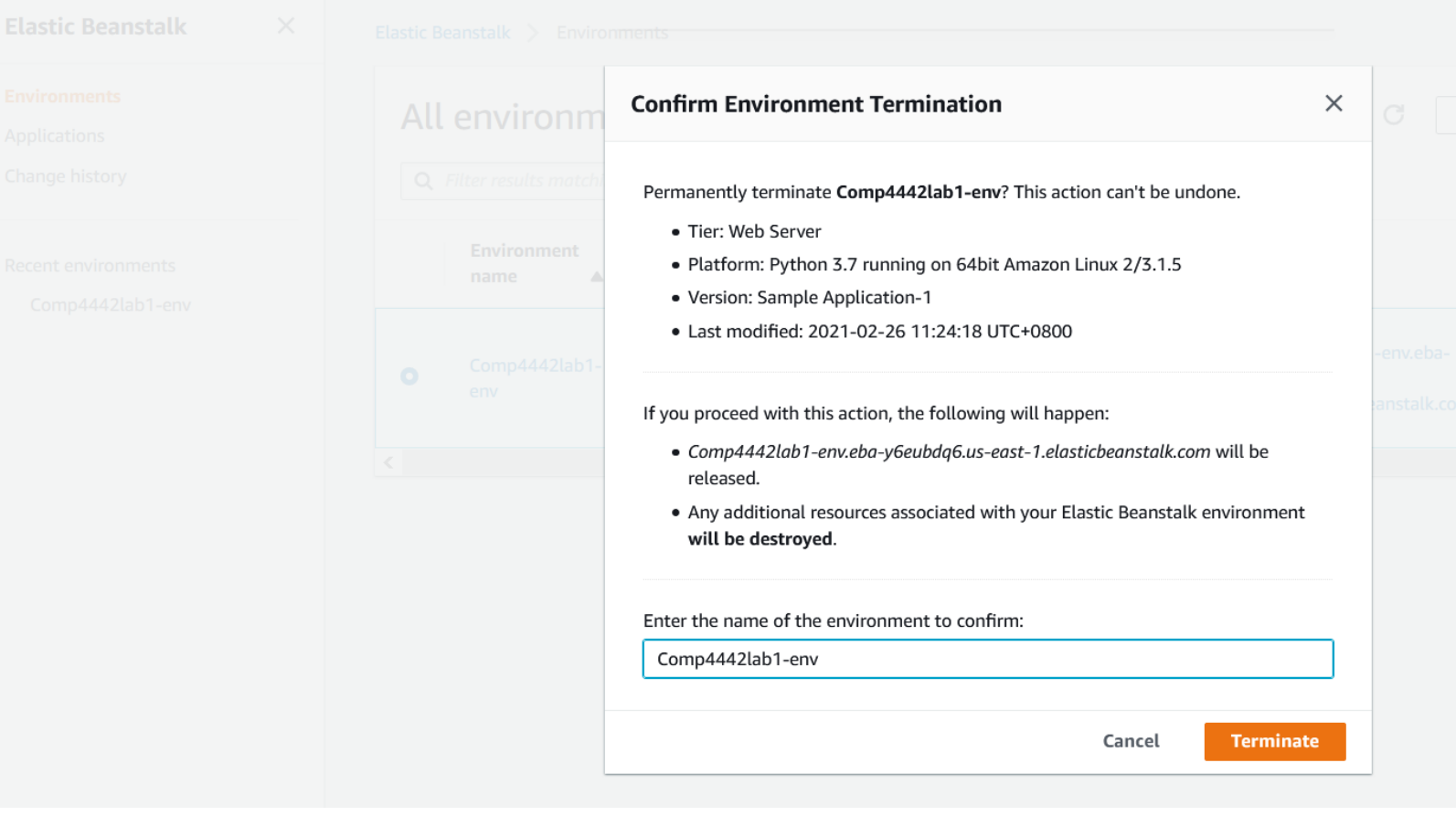
Load configuration
Save configuration
Swap environment URLs
Clone environment
Abort current operation
Restart app server(s)
Rebuild environment
3 Terminate environment

1

Environment name	Health	Application name	Date created	Last modified	URL	Platform	Platfo state
Comp4442lab1-env	Ok	COMP4442-lab1	2021-02-26 11:12:43 UTC+0800	2021-02-26 11:24:18 UTC+0800	Comp4442lab1-env y6eubdq6.us-east-1.elasticbeanst	Python 3.7 running on 64bit Amazon Linux 2	Suppo

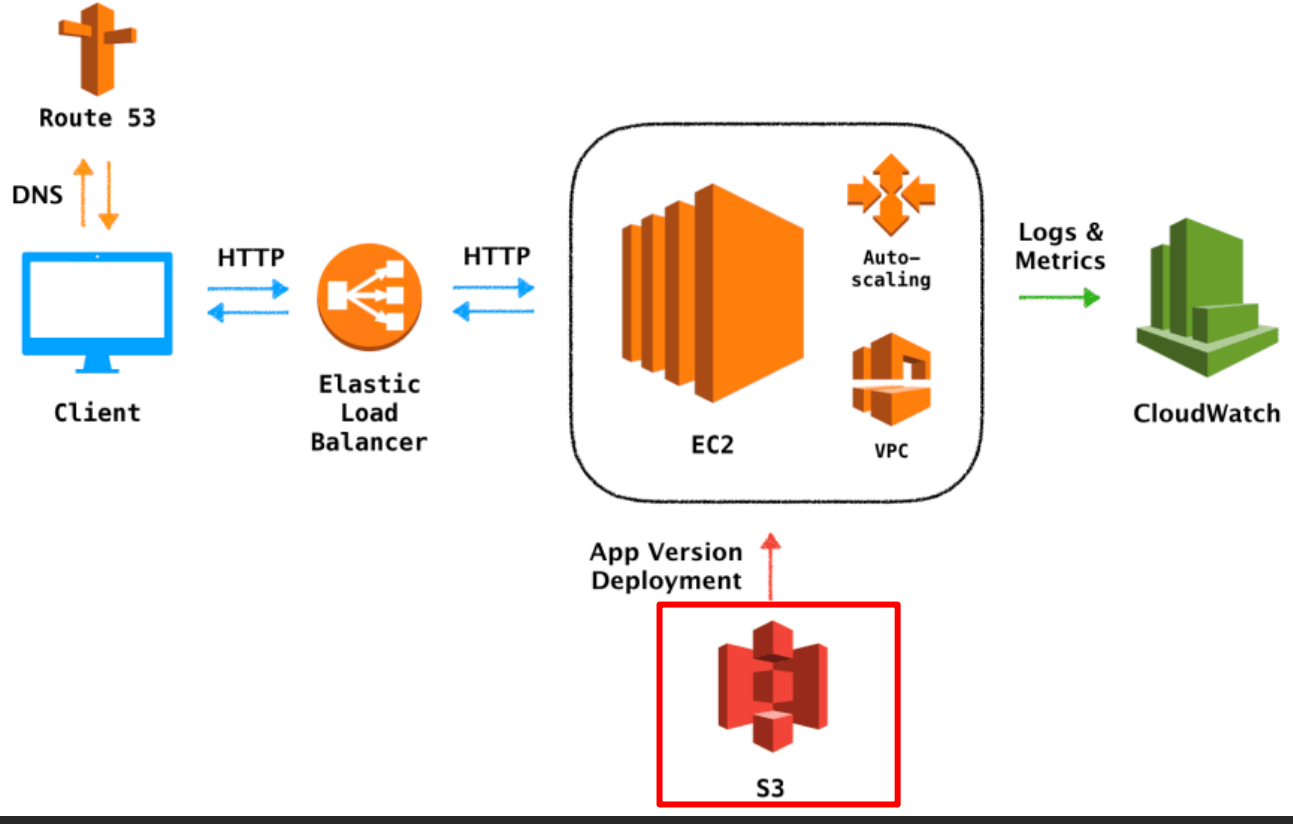
Stop Elastic Beanstalk

- Confirm your operation



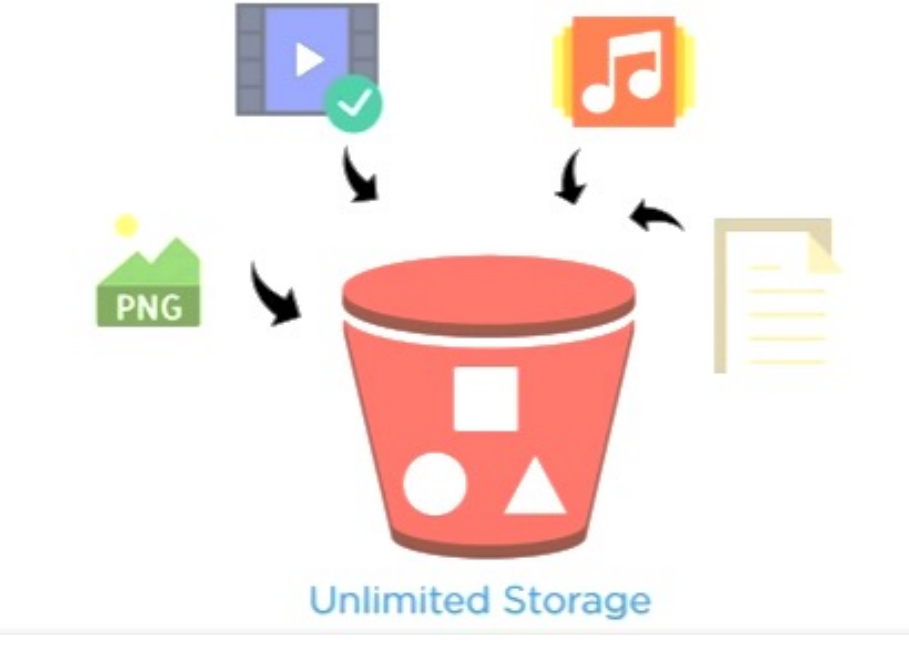
Review of Lab-01

- Elastic Beanstalk is not a single service
  - AWS Simple Storage Service (S3) is used to store the source code, web page, and media files, etc.



What is AWS S3

- AWS S3 provides object storage which is build for storing and recovering any amount of information or data from anywhere over the Internet.



What is AWS S3

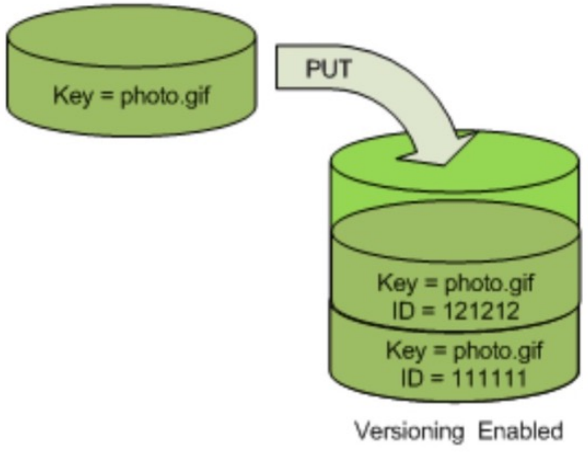
- Amazon S3 provides cloud storage through web services interface
- It provides 99.999999999% durability and 99.99% availability of objects
- It can store computer files up to 5 TB in size

Benefits of AWS S3

Basic concepts in S3

- An object consists of data, key (assigned name) and metadata
- A bucket stores objects
- When data is added to the bucket, S3 creates a unique version ID and allocates it to the object

For Example:

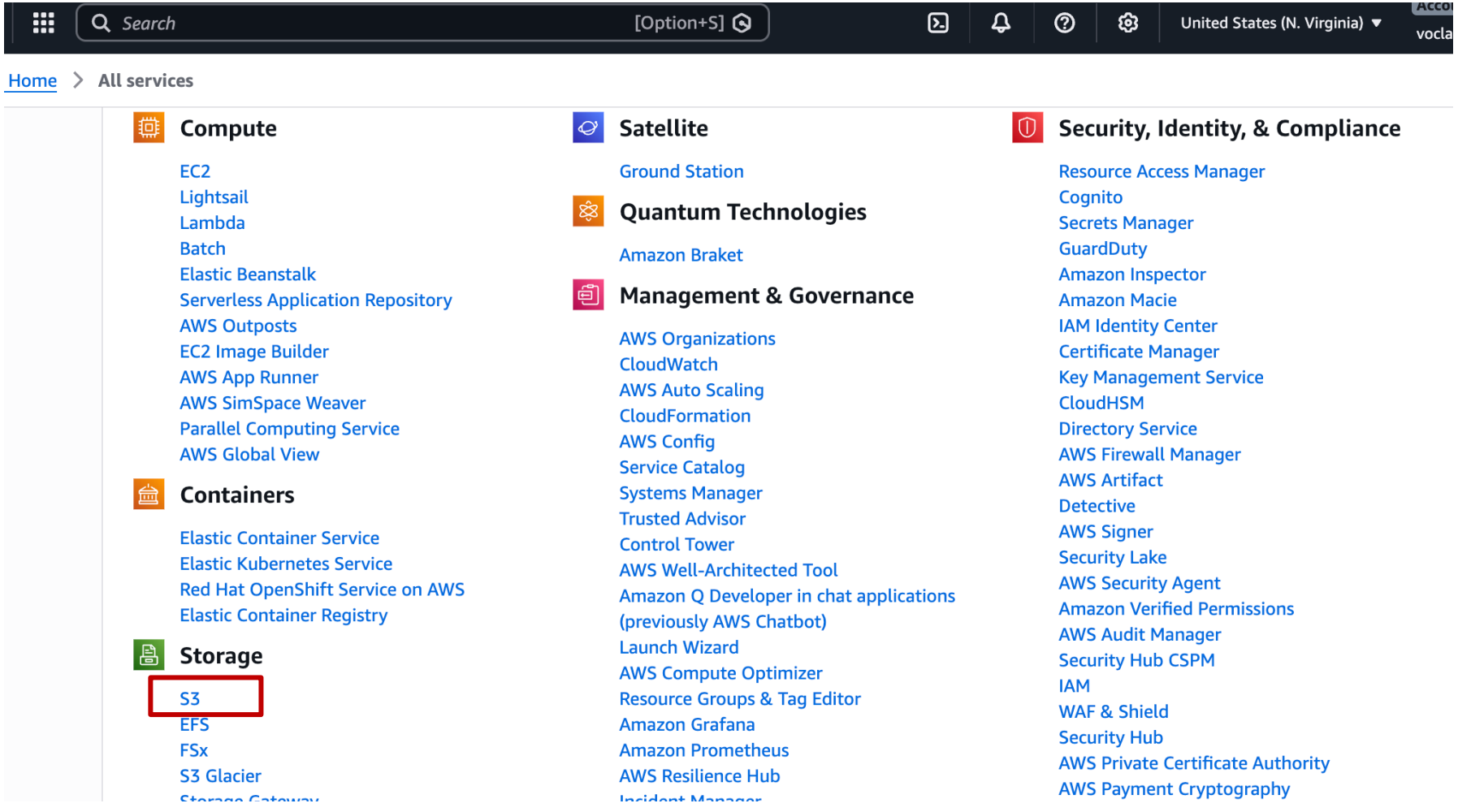


How to Use AWS S3

- Using the S3 console ✓
- Using the AWS Command Line Interface (CLI) ✓
- Using the AWS SDKs

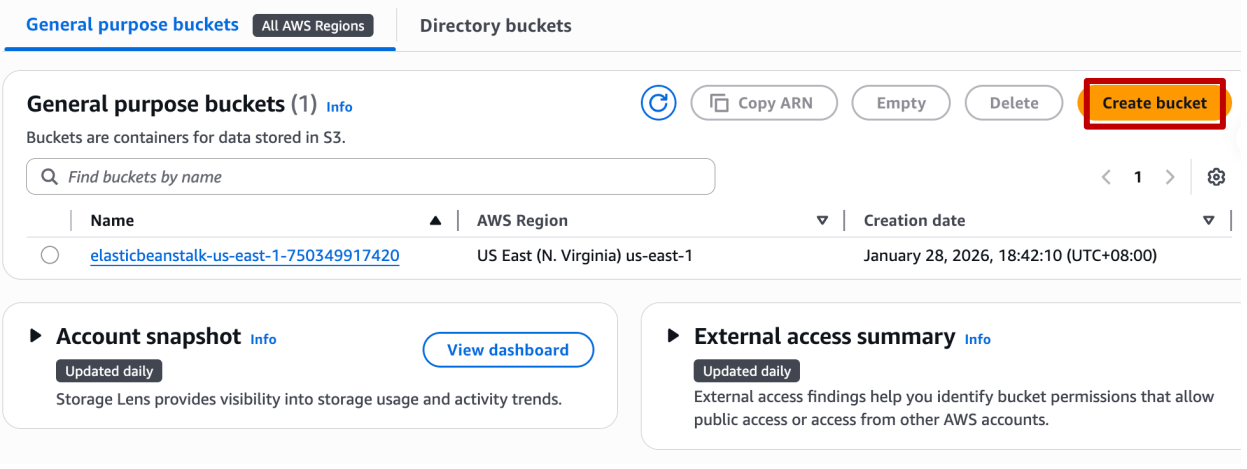
Use S3 with AWS Console

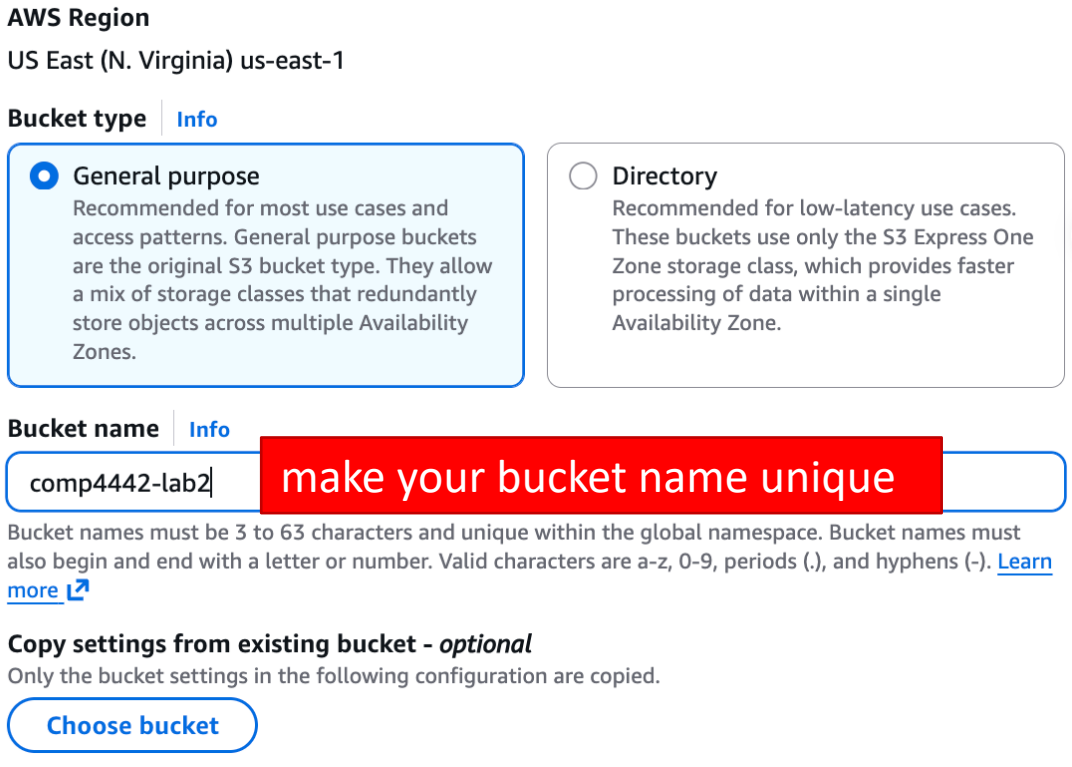
- Access the AWS S3 service



Use S3 with AWS Console

- Create bucket
  - Fill in the bucket name
  - Choose the region to store the bucket





Use S3 with AWS Console

Object Ownership

- ACLs disabled (recommended)
  All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
- ACLs enabled
  Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠ We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

Object Ownership

- Bucket owner preferred
  If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.
- Object writer
  The object writer remains the object owner.

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more

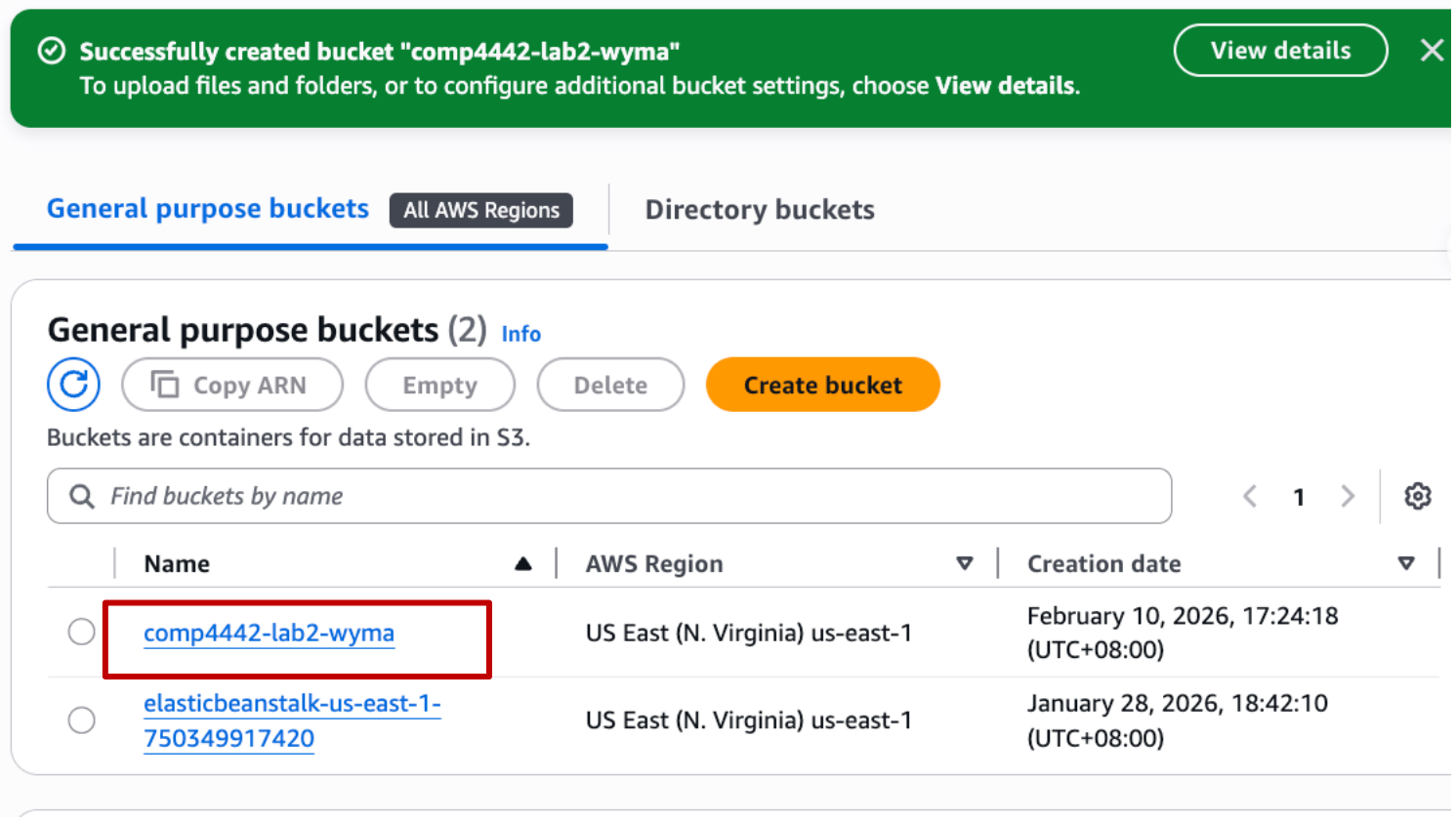
- [ ] **Block *all* public access**
  Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
  - [ ] **Block public access to buckets and objects granted through *new* access control lists (ACLs)**
    S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
  - [ ] **Block public access to buckets and objects granted through *any* access control lists (ACLs)**
    S3 will ignore all ACLs that grant public access to buckets and objects.
  - [ ] **Block public access to buckets and objects granted through *new* public bucket or access point policies**
    S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
  - [ ] **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**
    S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

⚠ **Turning off block all public access might result in this bucket and the objects within becoming public**
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

- [x] I acknowledge that the current settings might result in this bucket and the objects within becoming public.

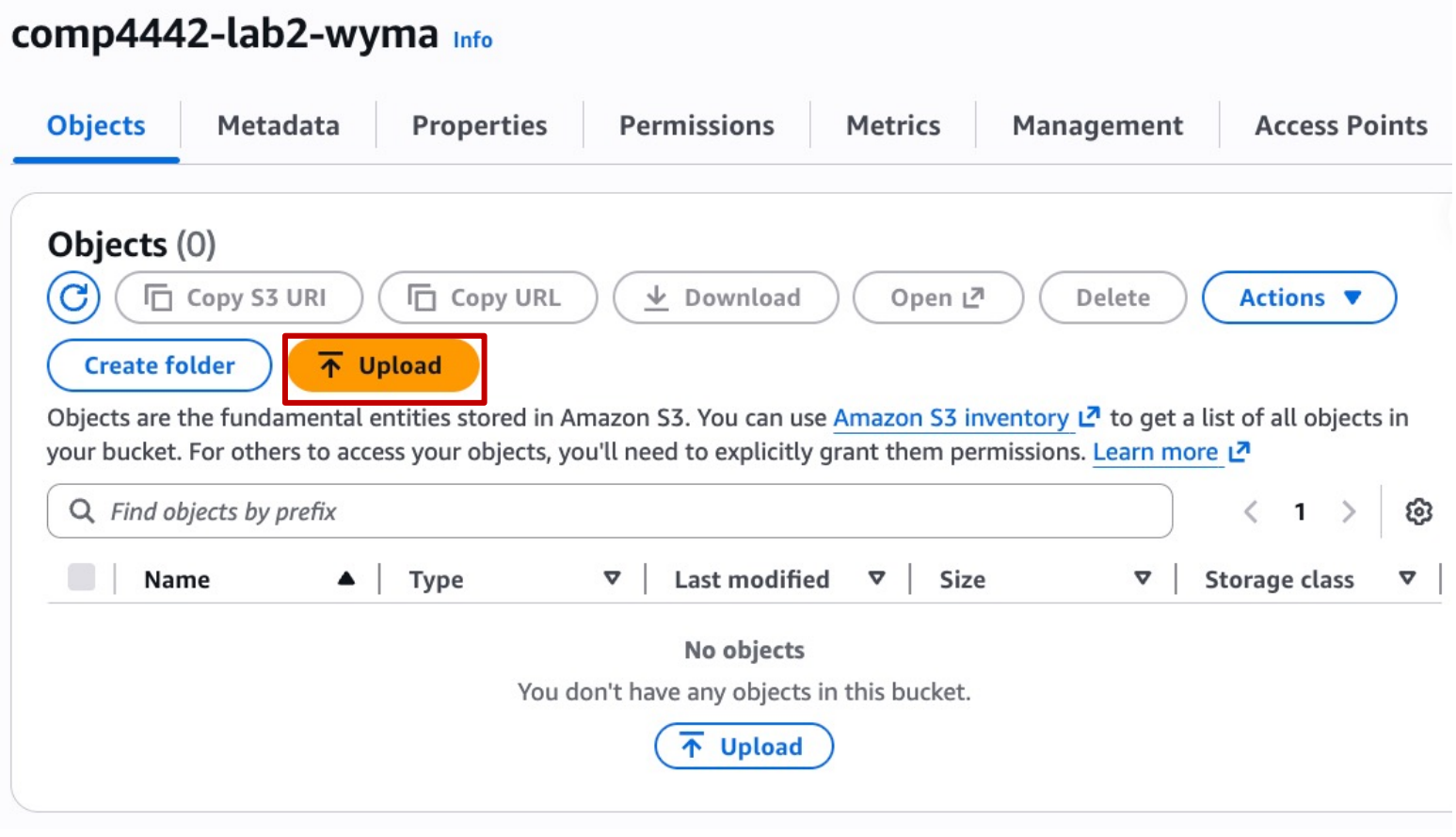
Use S3 with AWS Console

- Upload objects



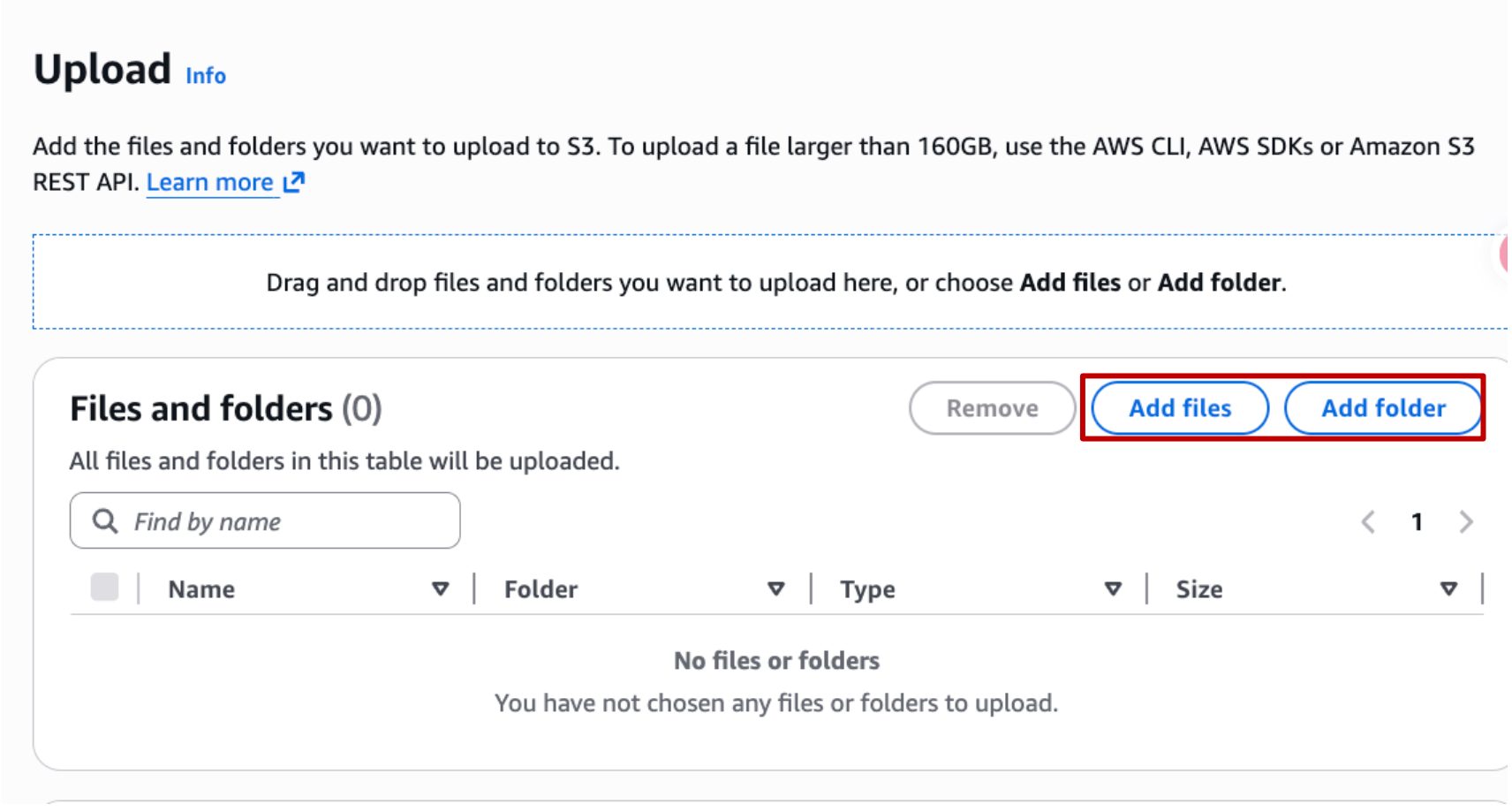
Use S3 with AWS Console

- Upload objects



Use S3 with AWS Console

- Upload objects



Use S3 with AWS Console

- Set permission

Files and folders (1 total, 25.6 KB)

Remove | Add files

All files and folders in this table will be uploaded.

Find by name

Name	Folder	Type	Size
S3-logo.png	-	image/png	25.6 KB

Destination Info

Destination

s3://comp4442-lab2-wyma

Destination details

Bucket settings that impact new objects stored in the specified destination.

Permissions

Grant public access and access to other AWS accounts.

Properties

Specify storage class, encryption settings, tags, and more.

Permissions

Grant public access and access to other AWS accounts.

Access control list (ACL)

Grant basic read/write permissions to other AWS accounts. Learn more

AWS recommends using S3 bucket policies or IAM policies for access control. Learn more

Access control list (ACL)

- Choose from predefined ACLs
- Specify individual ACL permissions

Grantee	Objects	Object ACL
Object owner (your AWS account) Canonical ID: dbddfc6402090e4866586071f5cff7e29b3110c2abe3d9262ef0aa3766ae5717	☑ Read	☑ Read ☑ Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	☑ Read	☐ Read ☐ Write
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	☐ Read	☐ Read ☐ Write

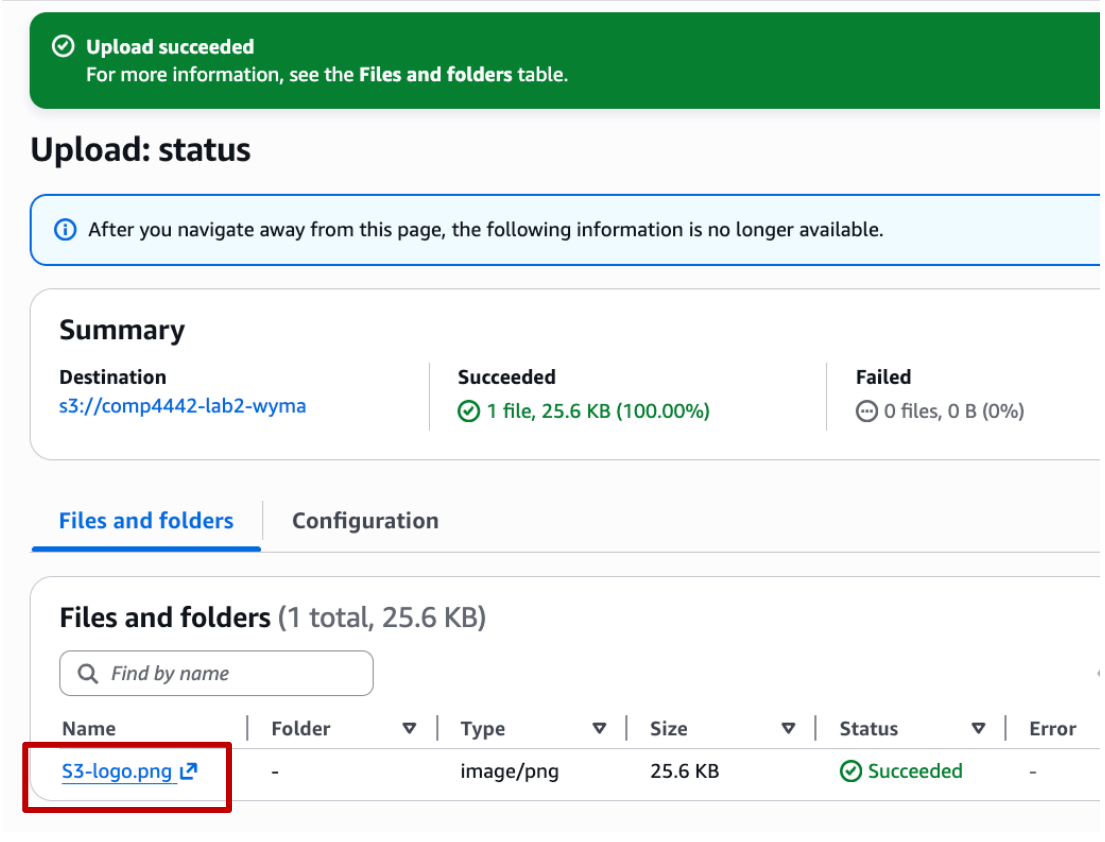
When you grant access to the Everyone or Authenticated users group grantees, anyone in the world can access the specified objects.

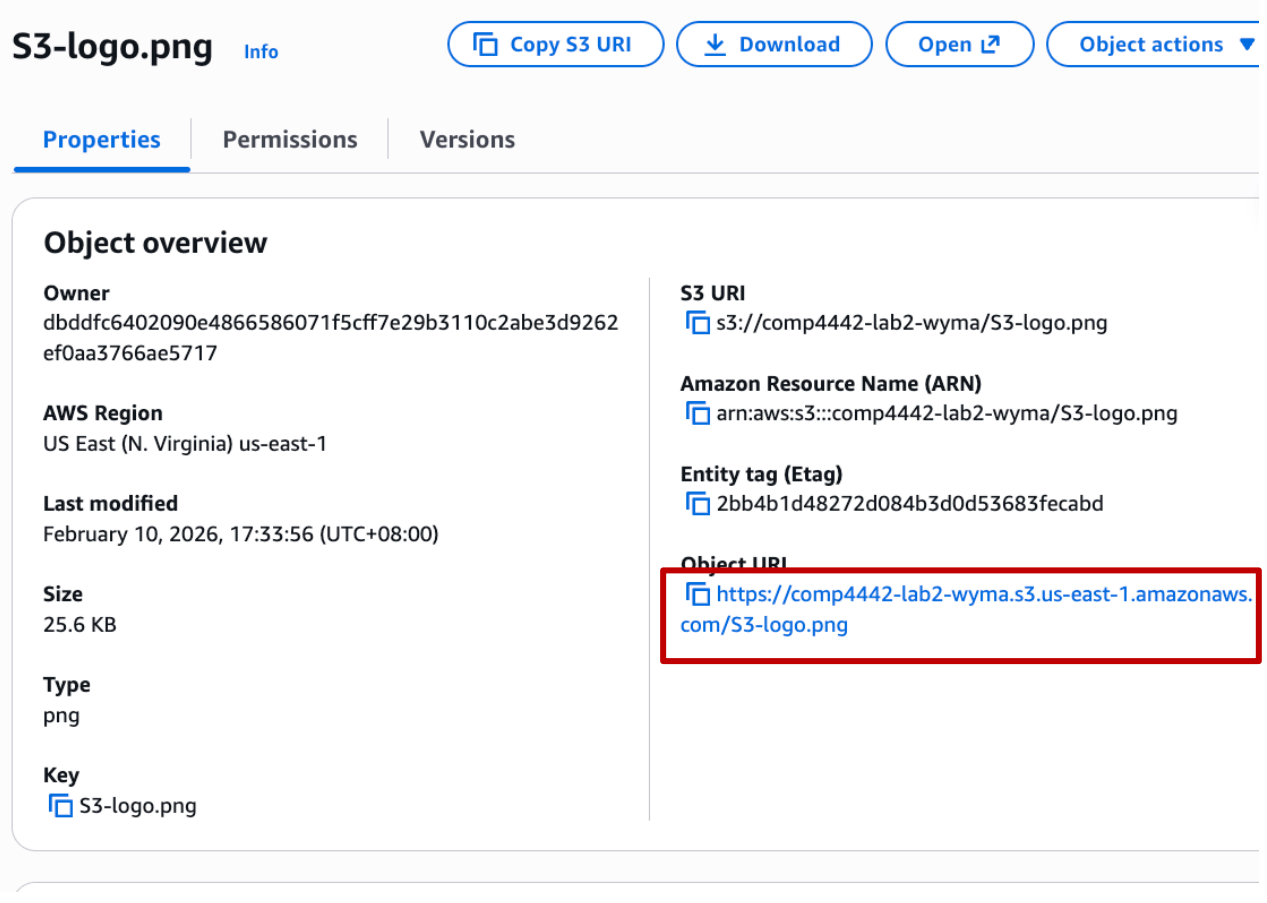
Learn more

☑ I understand the effects of these changes on the specified objects.

Use S3 with AWS Console

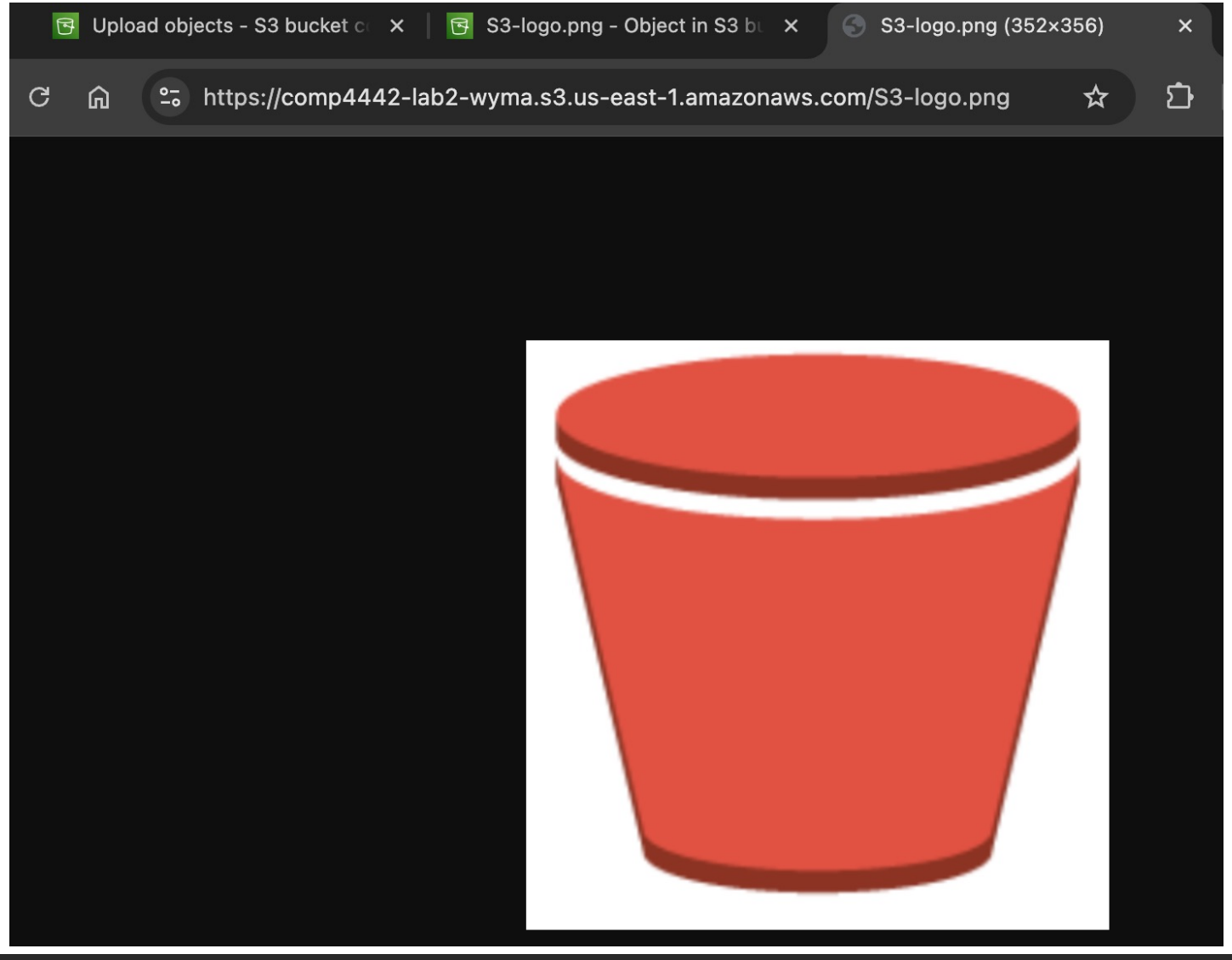
- Get Objects via URL
  - Click the object and get the URL





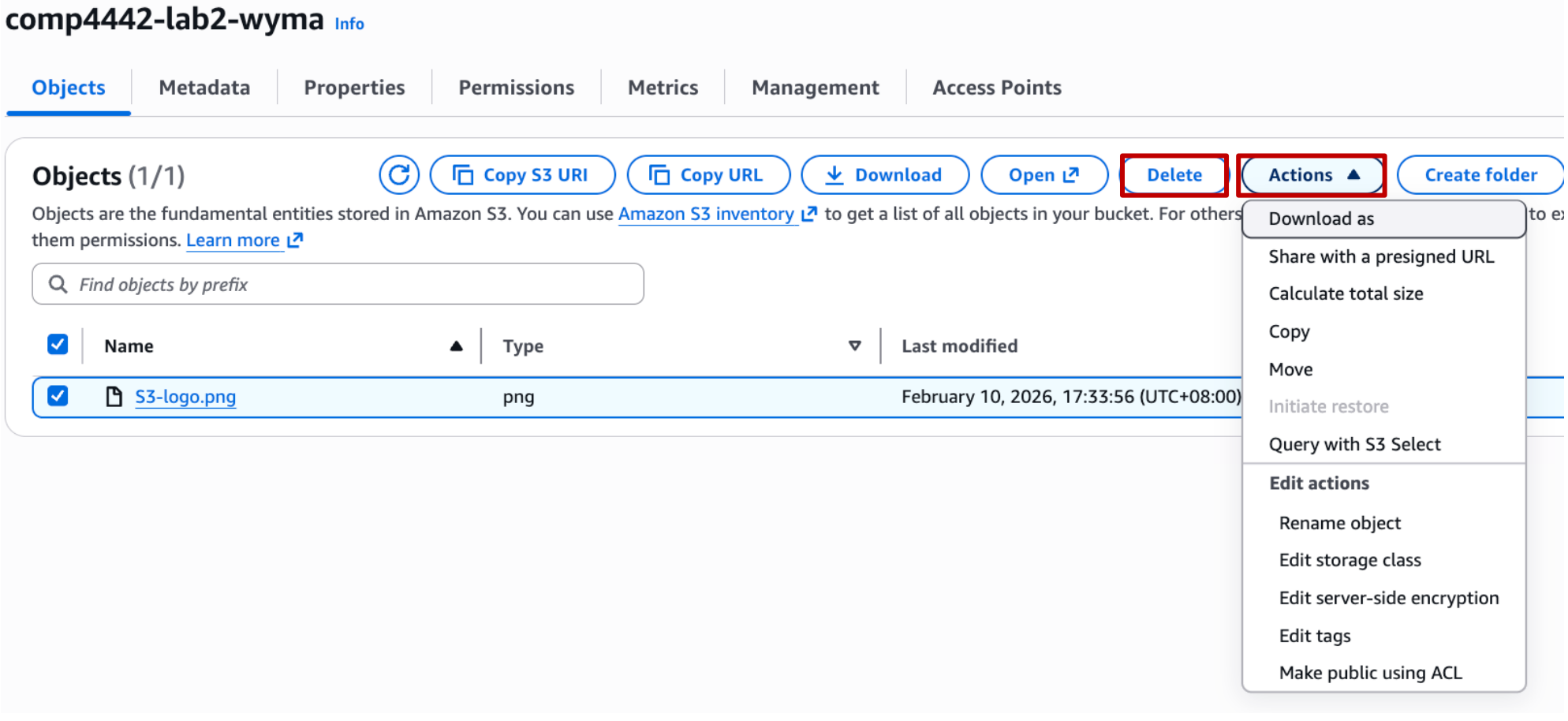
Use S3 with AWS Console

- Get Objects via URL



Use S3 with AWS Console

- Delete and Actions



Use S3 with AWS Console

- Delete bucket and objects

Delete objects Info

- If a folder is selected for deletion, all objects in the folder will be deleted, and any new objects added while the delete action is in progress might also be deleted. If an object is selected for deletion, any new objects with the same name that are uploaded before the delete action is completed will also be deleted.
- Deleting the specified objects can't be undone.

Learn more

Specified objects

Find objects by name

Name	Type	Last modified	Size
S3-logo.png	png	February 10, 2026, 17:47:24 (UTC+08:00)	25.6 KB

Permanently delete objects?

To confirm deletion, type *permanently delete* in the text input field.

permanently delete

Cancel | Delete objects

General purpose buckets All AWS Regions | Directory buckets

General purpose buckets (1/2) Info

Copy ARN | Empty | Delete | Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

1

Name	AWS Region	Creation date
comp4442-lab2-wyma	US East (N. Virginia) us-east-1	February 10, 2026, 17:24:18 (UTC+08:00)
elasticbeanstalk-us-east-1-750349917420	US East (N. Virginia) us-east-1	January 28, 2026, 18:42:10 (UTC+08:00)

Delete bucket Info

- Deleting a bucket cannot be undone.
- Bucket names are unique. If you delete a bucket, another AWS user can use the name.
- If this bucket is used with a Multi-Region Access Point in an external account, initiate failover before deleting the bucket.
- If this bucket is used with an access point in an external account, the requests made through those access points will fail after you delete this bucket.

Learn more

Delete bucket "comp4442-lab2-wyma"?

To confirm deletion, enter the name of the bucket in the text input field.

comp4442-lab2-wyma

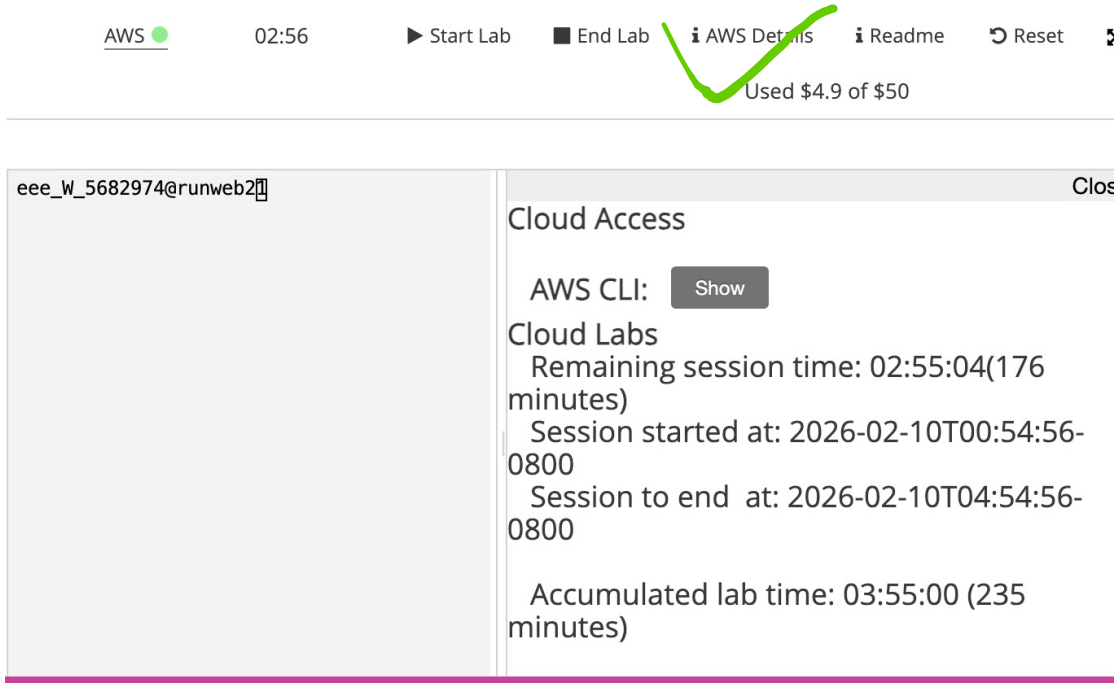
Use S3 with AWS CLI

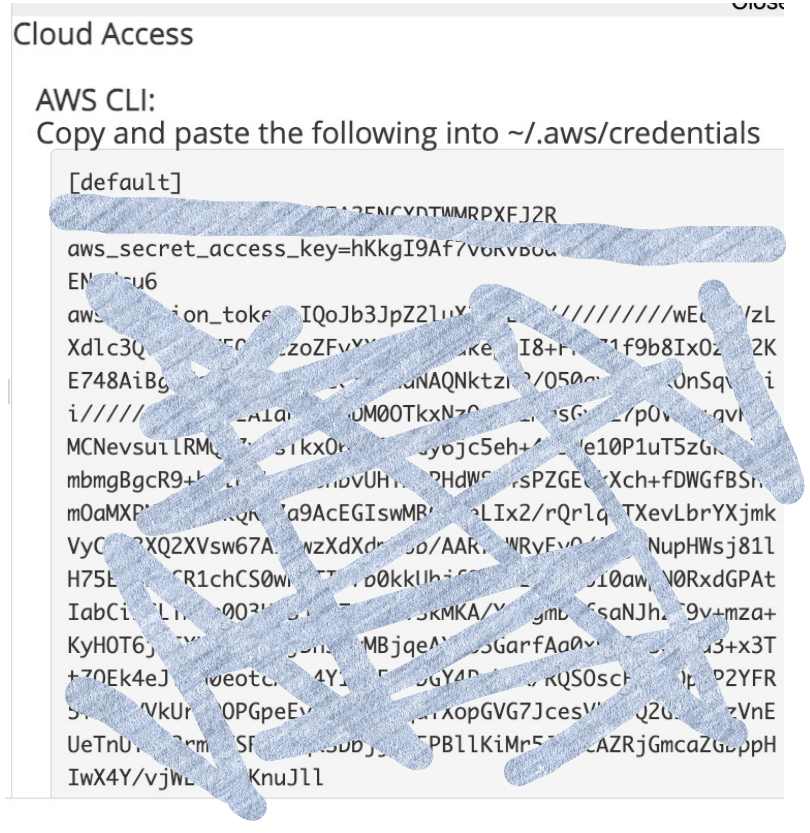
- What is AWS CLI?
  - An opensource tool that enables you to interact with AWS services using commands in your command-line shell.
  - Windows: https://awscli.amazonaws.com/AWSCLIV2.msi
  - MacOS: https://awscli.amazonaws.com/AWSCLIV2.pkg

- Why AWS CLI
  - The console cannot upload files larger than 160GB
  - Console may not be suitable for development

Use S3 with AWS CLI

- Preparation
  - Install the AWS CLI (check the installation files in Blackboard or by URL)
  - Check the account details and get the credentials





Use S3 with AWS CLI

- Configure the CLI(details can be found in Lab-02 Manual.pdf)
  - Copy the contents of your credentials to `C:\Users\<user_name>\.aws` （macOS users: ~/.aws/credentials）
  - Access the CLI in windows command line interface

```
C:\Users\ZHANG>aws configure
AWS Access Key ID [****************V7SM]:
AWS Secret Access Key [****************pjkZ]:
Default region name [us-east-1]:
Default output format [json]:

C:\Users\ZHANG>aws s3 mb s3://comp4442-lab2
make_bucket: comp4442-lab2

C:\Users\ZHANG>aws --version
aws-cli/2.1.27 Python/3.7.9 Windows/10 exe/AMD64 prompt/off

C:\Users\ZHANG>_
```

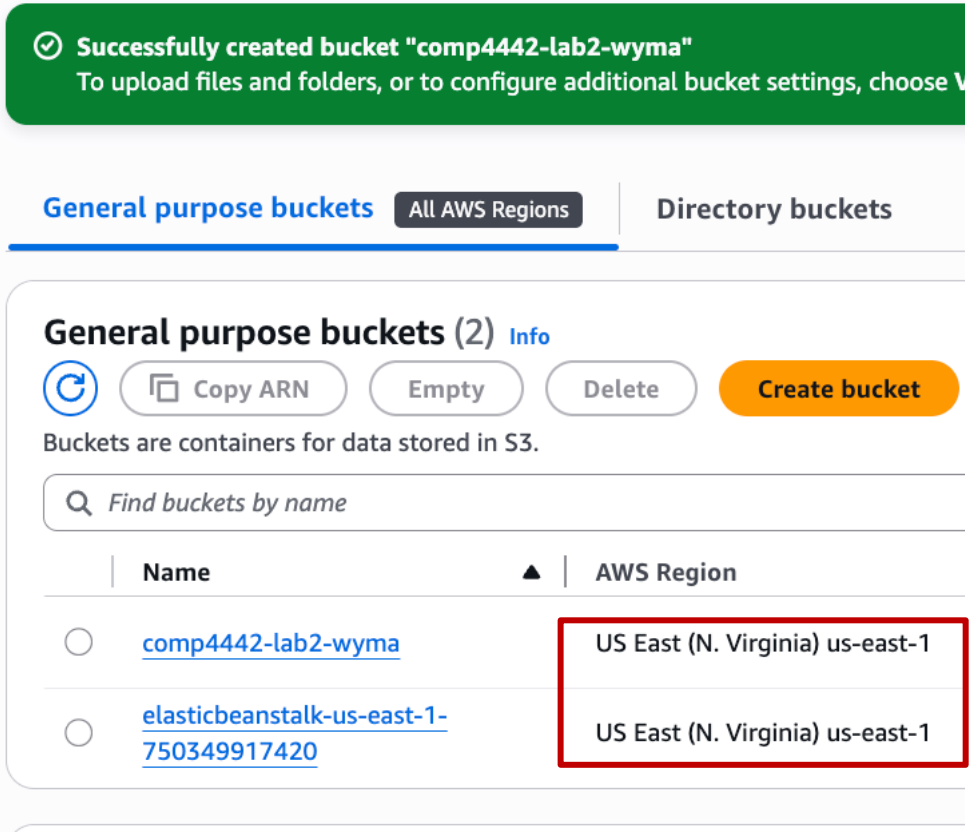
Use S3 with AWS CLI

- A detailed manual is in “Lab-02 Manual.pdf”

Use S3 with AWS CLI

- Check your Region
  - `aws configure get region`
- Set your Region if it does not match
  - `aws configure set default.region us-east-1`

  OR
  - `export AWS_DEFAULT_REGION=us-east-1`




```
[→ Lab2 % aws configure set default.region us-east-1
[→ Lab2 % aws configure get region
 us-east-1
```

Use S3 with AWS CLI

- Create bucket
  - `aws s3 mb s3://bucket-name`
- List buckets and objects
  - `aws s3 ls`
  - `aws s3 ls s3://bucket-name`
- Upload objects
  - `aws s3 cp filename.txt s3://bucket-name`
- Delete buckets
  - `aws s3 rb s3://bucket-name`
  - `aws s3 rb s3://bucket-name --force`

Use S3 with AWS CLI

- Delete objects
  - `aws s3 rm s3://bucket-name/example --recursive`
- Set permission
  - aws s3api put-object-acl --bucket comp4442 --key S3-logo.PNG --acl public-read
- For more information, refer to
  - https://awscli.amazonaws.com/v2/documentation/api/latest/reference/s3/index.html
  - https://awscli.amazonaws.com/v2/documentation/api/latest/reference/s3api/index.html

Q&A